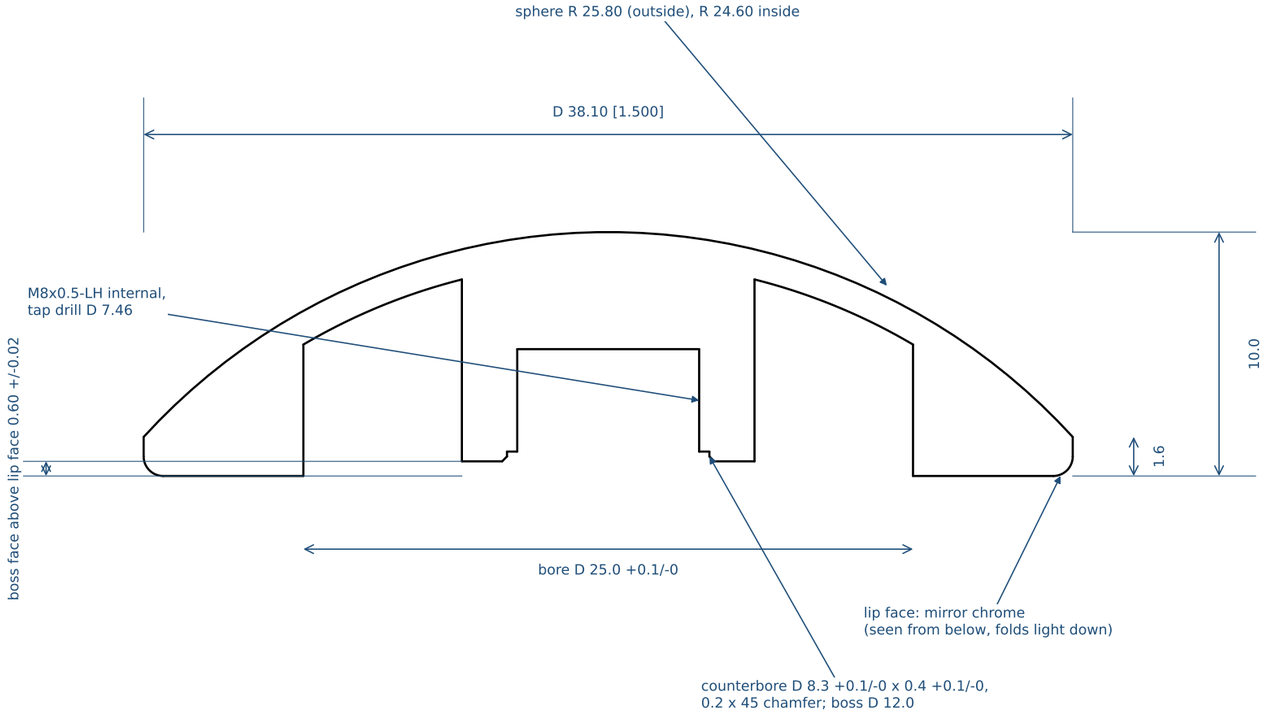
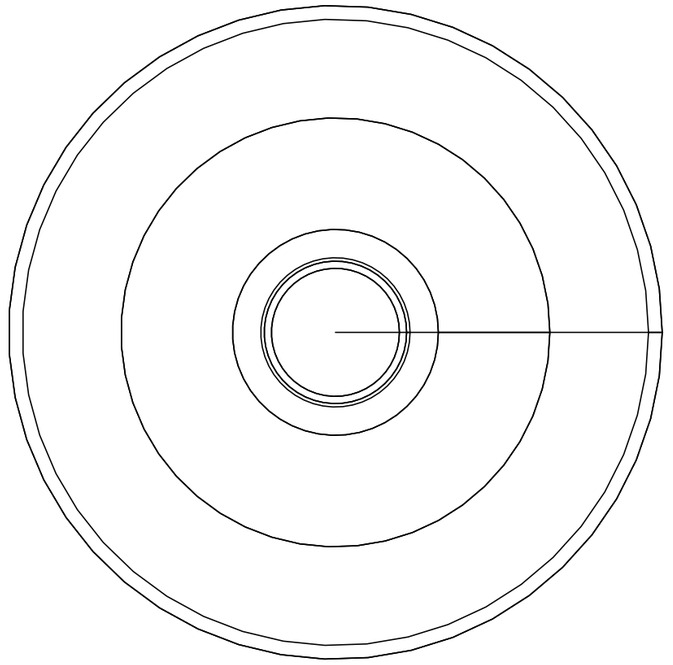
SECTION THROUGH THE AXIS (y=0); z = cap frame, contact face of the core = 0



BOTTOM VIEW



NOTES:

1. Outside is Class A (inspect at 300 mm daylight, zero pits/haze). The lip bottom face is seen lit from below: polish and chrome it like the dome.
2. Sphere R 25.80, height 8.4 above a 1.6 band; lip edge R 0.8. Shell 1.2 thick at the axis.
3. Centre boss D 12.0: internal M8x0.5 LEFT-HAND, tap drill D 7.46 (the STEP shows the tap-drill bore), full thread to 4.6 from the boss face. Boss MOUTH: counterbore D 8.3 +0.1/-0 x 0.4 +0.1/-0 deep with a 0.2 x 45 chamfer. It clears the unthreaded D 8.0 key on the core, so the boss can only ever land on the LED PCB.
4. THE BOSS FACE IS THE HARD STOP of this cap. Boss face flat and square to the thread within 0.03 (it clamps the LED PCB and is a return contact). DATUM: boss face stands 0.60 +/-0.02 ABOVE the lip face; lip face parallel to the boss face within 0.03.
5. Assembled, the lip face sits 0.25 above the louver and the window (stack of all tolerances 0.12): the lip must never touch them.
6. Mates: screws onto the RS-CAP-12L-CORE spigot (LH thread, Loctite 270 with activator 7649 on nickel, 1.5 N.m; keep adhesive off the boss face). LH so unscrewing the cap tightens this joint. Variants -CR / -BR / -BK.

RECORD SCONCE | LIT DOME CAP SHELL 1.5 in (label light)

PART NO: RS-CAP-12L-DOME REV: B DATE: 2026-11-11 SUPERSEDES: rev A, A.1 (do not build or quote)

MATERIAL: Brass C360 (CDA 360) bar, turned. Same outside profile as RS-CAP-12 (band + R 25.80 sphere segment).

FINISH: OUTSIDE (dome, band, lip bottom face r 18.05..19.05): polish to mirror Ra <= 0.1 um, then Cu + bright Ni + decorative Cr per ASTM B456 SC2. INSIDE (cavity, wall r 12.5, boss, thread): MASK, no chrome; bare or nickel. Rack on the inside only.

UNITS: mm [in] THIRD-ANGLE PROJECTION SCALE: as shown

GENERAL TOLERANCES: x.x +/-0.3 x.xx +/-0.1 angles +/-0.5 deg unless noted